

Equipment Lifts Averages, Not Equity: Multilevel Evidence from Taiwan's

TIMSS 2023



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Introduction

Billions are invested in school ICT infrastructure worldwide, promising both higher achievement and reduced inequality. This study provides design-correct evidence from Taiwan's first fully computer-based TIMSS (2023) on what such investments actually deliver in a high-connectivity system.

Research Questions:

1. Does school-level ICT raise mathematics performance?
2. Does ICT narrow inequalities by socioeconomic status (SES) and urban–rural location?

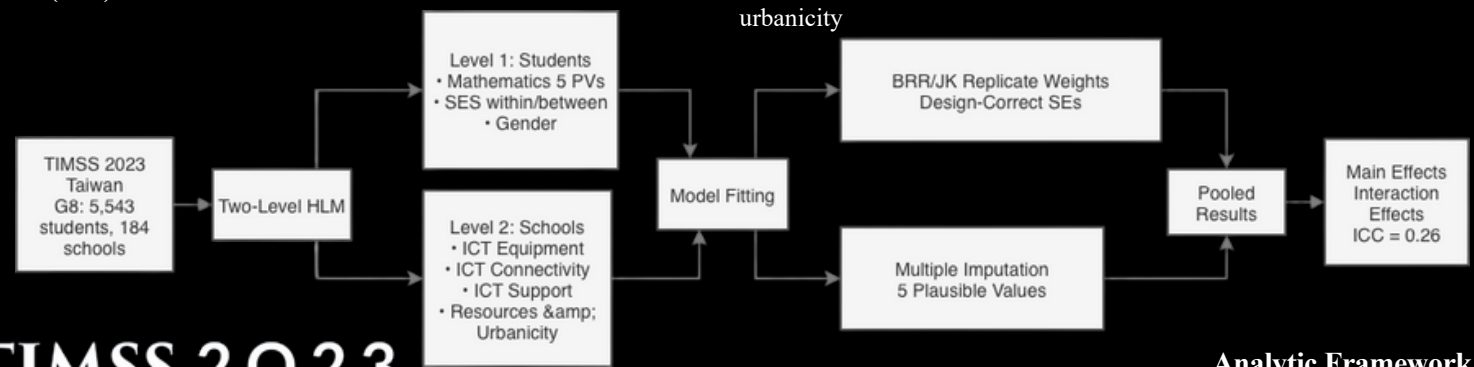
Method

Data: Taiwan Grade 8, TIMSS 2023

Design: **Two-level hierarchical linear models**

- Five plausible values with BRR/JK replicate-weight standard errors
- Group-mean–centered student SES
- Random intercepts (and tested random SES slope)
- Urbanicity indicators, school-mean SES, general school resources

ICT Components: Equipment (computers/devices) 、 Connectivity (internet access)、 Support (technical staff/training)
Controls: Student SES (within/between), gender, school resources, urbanicity



TIMSS 2023

Analytic Framework

Key Finding

FINDING 1: Equipment Works—Modestly

+1 SD increase in school equipment = +8–10 TIMSS points (≈ 0.08 – 0.10 SD)

Connectivity and support show no reliable direct associations

FINDING 2: ICT is NOT an Equalizer

SES $\times$ ICT interaction: Non-significant

Simple slopes at ± 1 SD ICT: Small and overlapping
→ No meaningful flattening of SES gradient

FINDING 3: Geographic Inequality Persists

Urban–rural gaps remain large: -27 to -53 points (≈ 0.27 – 0.50 SD)

ICT $\times$ Urbanicity interactions: Non-significant

FINDING 4: Substantial School Clustering

ICC ≈ 0.26 (26% of variance between schools)

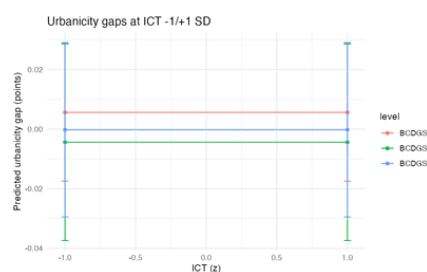
Adding covariates reduces between-school variance, but not via ICT interactions

TABLE 1: Main Effects Summary Table

Points are TIMSS scale points (mean ≈ 500 , SD ≈ 100). * $p < 0.05$

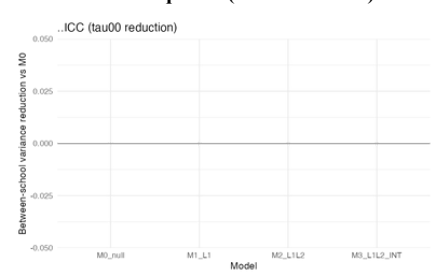
Predictor	Estimate	95% CI	Significant?
ICT Equipment (z)	+9.2	[+3.5, +14.8]	Yes*
ICT Connectivity (z)	+0.6	[-2.1, +3.2]	No
ICT Support (z)	+0.4	[-2.4, +3.1]	No
SES $\times$ ICT Equipment	-0.2	[-1.3, +0.9]	No
Urbanicity: Township vs Urban	-30.7	[-46.7, -14.6]	Yes*
Urbanicity: Rural vs Urban	-55.3	[-75.2, -35.4]	Yes*
Urbanicity: Remote vs Urban	-30.8	[-51.3, -10.3]	Yes*

FIG 1: Urbanicity Gaps by ICT Level



Urbanicity achievement gaps at ICT $-1/+1$ SD. Gaps remain large and do not vary meaningfully by ICT level, indicating ICT does not moderate geographic inequality.

FIG 2: Model Sequence (ICC Reduction)



Between-school variance reduction across model sequence (M0–M3). Adding covariates reduces clustering, but ICT interactions do not contribute meaningfully

Implications for Policy

Hardware elevates average performance but does not close existing achievement gaps. In high-connectivity systems like Taiwan's, the returns to infrastructure depend on how well it is integrated into instruction. Policy should therefore pair equipment upgrades with teacher professional development, formative and data-informed practices, targeted student supports, and an explicit equity-focused design.

Limitation, Future work & Conclusion

- Cross-sectional design (causal inference limited)
- Potential unobserved school-level confounding
- Future: Multi-cycle harmonized measures or policy shocks for stronger causal inference

→ **Hardware helps averages; equity requires more than hardware.**

Reference

Von Davier, M., Kennedy, A., Reynolds, K., Fishbein, B., Khorramdel, L., Aldrich, C., Bookbinder, A., Bezirhan, U., & Yin, L. (2024). TIMSS 2023 International Results in Mathematics and Science. <https://doi.org/10.6017/lsc.tpisc.timss.rs6460>

Wang, Y., & Wang, Y. (2023). Exploring the relationship between educational ICT resources, student engagement, and academic performance: A multilevel structural equation analysis based on PISA 2018 data. *Studies in Educational Evaluation*, 79, 101308. <https://doi.org/10.1016/j.stueduc.2023.101308>

Huang, F. L. (2024). Using plausible values when fitting multilevel models with large-scale assessment data using R. *Large-scale Assessments in Education*, 12(1). <https://doi.org/10.1186/s40536-024-00192-0>